

Problem Set 4

Spring 2026

Due: 11:59 pm Monday, 4 May

To facilitate grading and timely feedback
please note that all submissions are through Gradescope.

Solve each problem on a new page.

Please clearly identify collaborator names and the resources used for each problem.

1. Assume that the formal system F is sound, that the axioms and deduction rules of F can all be given explicitly, and that F is capable of expressing any ordinary mathematical reasoning, for example, about Turing machines and proofs. In other words, F is simply one of the usual systems to which Gödel's Incompleteness Theorem applies.

Let $L = \{x \mid x \text{ encodes a proof, in } F, \text{ that } 1 + 1 = 3\}$.

- (a) Is L the empty language?
- (b) Is your answer to (a) provable in F ?
- (c) Is L decidable?
- (d) Is your answer to (c) provable in F ?
- (e) Describe a Turing machine M , such that the statement:
' M runs in $O(2^n)$ time, where n is the length of M 's input'
is independent of F (i.e., neither provable nor disprovable in F).

2. Show that for all functions $f : \mathbb{N} \mapsto \mathbb{N}$, we have $\text{SPACE}(f(n)) \subseteq \text{TIME}(2^{O(f(n))})$.
3. Let G be an undirected graph with n vertices. Then a *Hamilton path* is a simple path in G that visits each vertex once (i.e., has n vertices and $n - 1$ edges), while a *Hamilton cycle* is a simple cycle in G that visits each vertex once (i.e., has n vertices and n edges). Let *HAMPATH* and *HAMCYCLE* be the problems of deciding whether G has a Hamilton path and Hamilton cycle respectively, given G as input.
 - (a) Show that if G has a Hamilton cycle, then G also has a Hamilton path.
 - (b) Give an example of a graph G that has a Hamilton path but no Hamilton cycle.
 - (c) Give a polynomial-time reduction from *HAMCYCLE* to *HAMPATH*.
 - (d) Give a polynomial-time reduction from *HAMPATH* to *HAMCYCLE*.
4. In the *quadratic programming* (*QUADPROG*) problem, the input is a system of equalities and inequalities, each involving polynomials of degree at most 2 (with integer coefficients) in n real variables $x_1, \dots, x_n$. The problem is to decide whether there exists an assignment to $x_1, \dots, x_n$ that satisfies all the constraints simultaneously. As an example, the system

$$\begin{aligned} x_1 + x_2 &\leq 1 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \\ 4x_1x_2 &\geq 1 \end{aligned}$$

can be satisfied by setting $x_1 = x_2 = 1/2$, but if we replaced the last inequality by $x_1x_2 \geq 1$, then the system would be unsatisfiable.

- (a) Show that *QUADPROG* is NP-hard, by reduction from any problem that was already proved NP-hard in class.
 - (b) What is a difficulty in showing that *QUADPROG* $\in$ NP?
5. We have $\text{EXP} = \bigcup_k \text{TIME}(2^{n^k})$, and that $\text{NEXP} = \bigcup_k \text{NTIME}(2^{n^k})$. Just as it is open problem whether $\text{P} = \text{NP}$, it is also an open problem whether $\text{EXP} = \text{NEXP}$. Show that if $\text{P} = \text{NP}$, then $\text{EXP} = \text{NEXP}$ also. (*Hint*: Given a language $L \in \text{NEXP}$, can you come up with a modified language $L' \in \text{NP}$, such that $L \in \text{EXP}$ if and only if $L' \in \text{P}$?)